PORTFOLIO · 2026

Home LLM Inference Platform

Running large language models on Kubernetes,
from the GPU up to the application

Toshi

github.com/htatemura

WHY THIS PROJECT

Proof that I can run LLMs in production, not just call an API.

Close the gap

28 years of infrastructure, support and PM work, but no LLM job title yet. This project supplies the evidence.

Cover both layers

Understand the model, and operate the GPU platform underneath it. Few candidates can speak to both.

Measure, not just build

Design choices are backed by monitoring data and benchmarks, and written up publicly.

About me

28 years
L3 support, professional services, robotics engineering, marketing

10+ years
Program and project management

JP / EN
Bilingual; 3 years working in Singapore

CURRENT ROLE

American semiconductor equipment company, Japan. Founding employee of a German software company's Japan subsidiary, later acquired. Semiconductor process optimization software.

DISTRIBUTED SYSTEMS

Production experience with Apache Ignite and Microsoft Orleans

CERTIFICATIONS

NVIDIA NCA-AIIO · CNCF KCNA · CCNA · SCJP

Architecture

Solid = running today · Dashed = planned

APPLICATIONS	RAG over 729 recorded sessions	Personal small language model (nanoGPT-based)			
SERVING	Ollama on Kubernetes (CPU)	vLLM on GPU	Gateway API		
PLATFORM	kubeadm · 1 control + 2 workers	Cilium CNI	Harbor registry	local-path storage	NVIDIA device plugin
OBSERVABILITY	Prometheus	Grafana + alerting	DCGM exporter	vLLM metrics	
HARDWARE	HP EliteDesk 800 G6 · KVM	Synology NAS	RTX A2000 12 GB · passthrough to LLM node		

Where it stands today

Component	Status	Notes
kubeadm cluster	Running	1 control plane + 2 workers on KVM VMs
Cilium CNI	Running	Enforces standard NetworkPolicy
Harbor registry	Running	Self-hosted image source for the cluster
LLM serving on Kubernetes (CPU)	Running	Ollama + Llama 3.2 3B on a dedicated LLM node
Model storage	Running	local-path volume; models survive Pod restarts
Prometheus + Grafana	Running	Host metrics and custom memory alerting
GPU worker	Running	Power supply replaced; RTX A2000 passed through to the LLM node
NVIDIA device plugin	Running	GPU schedulable as nvidia.com/gpu; verified with a test Pod
vLLM on GPU	Next	Same OpenAI-compatible API, swapped in on GPU
GPU and LLM dashboards	Planned	DCGM exporter and vLLM metrics

Roadmap in five phases

PHASE 1 · DONE

LLM serving on Kubernetes

Ollama on CPU with Harbor images and persistent model storage. OpenAI-compatible API verified.

PHASE 2 · NOW

GPU in the cluster

✓ New power supply
✓ RTX A2000 to the LLM node
✓ NVIDIA device plugin
vLLM on GPU
Gateway API

PHASE 3

Observability

DCGM for GPU health, vLLM metrics for latency and throughput, Grafana dashboards.

PHASE 4

Benchmarks

Compare quantization, concurrency and context length on a 12 GB GPU. Publish results.

PHASE 5

Applications

Run RAG over my recorded sessions and a personal small language model on top.

Phase 1: done · Phase 2: GPU live, vLLM next · Phases 2–3: about 1–2 months · Phase 4: about 1 month · Phase 5: ongoing

Questions the benchmarks will answer

Question	Variable	Metrics	Result
How many concurrent users fit on 12 GB?	Concurrent requests	Tokens/s, p95 latency	[TBD]
What does quantization trade away?	FP16 vs 4-bit	Tokens/s, VRAM, quality	[TBD]
How long can the context get?	Max model length	VRAM, time to first token	[TBD]
Which model size fits best?	Small vs mid-size models	Tokens/s, VRAM	[TBD]

Every result is published with its method, so anyone can reproduce it.

What broke, and how I fixed it

SYMPTOM

Long-running scripts killed whenever the remote desktop disconnected

ROOT CAUSE

systemd ended the user session (Linger=no)

FIX

loginctl enable-linger

SYMPTOM

Hardware video acceleration not working on a Haswell machine

ROOT CAUSE

Wrong VA-API driver for that GPU generation

FIX

i965 driver instead of iHD

SYMPTOM

Power error on the host after adding the GPU

ROOT CAUSE

Stock 260 W power supply

FIX

Power supply replaced; GPU reinstalled and scheduled from Kubernetes

What this demonstrates

Skill	Evidence
Kubernetes administration	kubeadm cluster with Cilium and Harbor; CNCF KCNA
GPU infrastructure	GPU passthrough on KVM, GPU scheduling via the NVIDIA device plugin; NVIDIA NCA-AIIO
LLM serving	Ollama on Kubernetes with an OpenAI-compatible API
Observability	Prometheus, Grafana and custom alerting in daily use
Troubleshooting	L3 support background; documented break-fix log
Communication	Write-ups published in Japanese and English

Next steps

CERTIFY

NVIDIA NCP-GENL for LLM depth. NCP-AIO added if targeting platform roles.

PUBLISH

Public GitHub repo, plus write-ups on Zenn (Japanese) and LinkedIn (English).

BUILD ON IT

Applications on the platform: RAG over recorded sessions and a personal SLM.

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github.com/htatemura · [blog URL]